PRANAV JINDAL

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EDUCATION

Bachelor of Technology in Computer Science and Engineering Graduated **June 2025**
Vellore Institute of Technology **8.22 CGPA**

TECHNICAL SKILLS

Programming: Python, C++, Java, SQL

Data Engineering: PySpark, ETL Pipelines, PostgreSQL, Data Validation, Batch Processing, Feature Engineering

ML & Computer Vision: Scikit-learn, XGBoost, CNN, YOLO, TensorFlow Lite, Depth Anything, GeoSAM

Backend & Systems: Linux, REST APIs, OOP, DBMS, Docker, Distributed Systems, Log Analysis

Cloud & DevOps: AWS Cloud Practitioner Certified, GCP, CI/CD Pipelines, Docker Compose

Networking & Telecom: Open5GS, OAI, UERANSIM, 5G Core, SBI, NAS, AMF, SMF, AUSF, UDM, NRF, UPF

Tools: Git, GitHub, VS Code, Wireshark, NVIDIA Jetson, Raspberry Pi, Rasterio, Shapely, Leafmap

WORK EXPERIENCE

Indian Institute of Technology Bombay, Mumbai: Project Research Assistant [On-Site] **October 2025 – Present**

- Working on SA 5G core network optimization using Open5GS, OAI, Linux-based testbeds, and SBI-based NF interactions.
- Introduced a custom NAS Anchor Function to offload AMF registration handling and reduce AMF processing load by 15%.
- Split the UE registration flow across NAF and AMF, handling AUSF interaction and security mode completion at NAF.
- Performed scalability testing with 512 parallel UEs, observing only a 0.25% increase in registration latency.

Indian Institute of Technology Roorkee, Roorkee: Project Research Associate [On-Site] **August 2025 – September 2025**

- Automated ADAS camera calibration using Depth Anything, reducing manual setup effort by 50%.
- Fine-tuned YOLO models on NVIDIA Jetson for real-time two-wheeler ADAS edge inference.
- Improved detection latency and reliability for ADAS deployment in resource-constrained embedded AI environments.
- Built Raspberry Pi telemetry pipelines for remote CPU, memory, and device-level monitoring.

Centre for Artificial Intelligence and Robotics, DRDO, Bangalore: Project Trainee [On-Site] **March 2025 – May 2025**

- Built a Python-based satellite image segmentation pipeline, reducing manual annotation effort by 60%.
- Implemented GeoSAM workflows to generate accurate masks and geospatial boundaries from remote sensing data.
- Designed CPU-compatible geospatial processing workflows, reducing dependency on GPUs and GIS-specific tools.
- Processed spatial outputs using Rasterio, Shapely, and Leafmap to support downstream geospatial analysis.

SVGS IT Solutions, Noida: Software Engineer Intern [Hybrid] **May 2024 – November 2024**

- Built Python-based machine learning workflows for data preprocessing, feature engineering, model training, and evaluation.
- Enhanced model performance by 15–20% through data cleaning, tuning, validation, and workflow refinement.
- Created dashboards to monitor business KPIs, model outputs, performance trends, and stakeholder insights.
- Automated recurring analytical reports using reusable Python workflows, reducing manual reporting effort.

PROJECTS

PySpark Financial Monitoring and Regulatory Alert Pipeline April 2026

Built a PySpark-based AML transaction monitoring pipeline to process transaction files, validate records, flag suspicious activity, and store alerts in PostgreSQL, simulating a BFSI-style workflow with regulatory-inspired controls and data quality checks.

- Built a PySpark batch ETL pipeline to extract, validate, transform, and load financial transaction data.
- Validated transaction records using Pydantic and separated invalid records into quarantined outputs.
- Implemented Spark DataFrame rules to flag high-value transfers and high-risk country transactions.
- Used Docker Compose with PostgreSQL to simulate a production-style financial data workflow.

Harvest Watch - AI-Based Plant Disease Detection April 2025

Built a mobile-ready plant disease detection system using CNN models, TensorFlow Lite, and an Android application for real-time farmer support.

- Trained a CNN model using InceptionV3 on the PlantVillage dataset with 54K+ plant disease images.
- Boosted model performance using preprocessing, data augmentation, hyperparameter tuning, and validation workflows.
- Converted the trained model to TensorFlow Lite and integrated it into an Android app for on-device inference.
- Evaluated model performance using accuracy metrics, test-time augmentation, and validation reports.

LEADERSHIP & ACTIVITIES

Data Science Club, VIT Bhopal, : Co-Lead | June 2023

- Led a 25-member team to organize 5+ technical events, improving student engagement by 70%.
- Mentored 20+ students in Python, machine learning fundamentals, and project-based learning.